

Carbon Accounting — Quick Reference

What does "Company emits X tCO₂e" actually mean?

It **does not mean** the company physically released that much carbon itself.

It means:

The activities required to run the company caused greenhouse gas emissions equivalent to X tons of CO₂.

Core Formula

Activity × Emission Factor = Emissions

Examples:

Electricity Used × Grid Factor

Diesel Burned × Diesel Factor

Plastic Purchased × Material Factor

The 3 Scopes

Scope 1 — Direct Emissions

Emissions generated by the company itself.

Examples:

- Diesel generator
- LPG boiler
- Natural gas furnace

Company burns fuel

↓

Company emits CO₂

Scope 2 — Purchased Energy

Emissions from bought electricity.

Example:

Factory uses electricity

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Power plant burns fuel

↓

Emissions assigned to factory

Scope 3 — Value Chain Emissions

Emissions embedded in purchased materials, transport, waste, etc.

Example:

Oil extraction

↓

Plastic manufacturing

↓

Transport

↓

Factory buys plastic

Those upstream emissions become part of the factory's footprint.

Why Virgin Plastic Dominates

A factory may only consume plastic pellets, but producing those pellets required:

Raw material extraction

↓

Refining

↓

Manufacturing

↓

Transport

All of those emissions are included in Scope 3.

That's why materials often dominate manufacturing footprints.

What is a Leak Point?

A leak point is **not a carbon leak**.

It is:

The activity contributing the largest share of total emissions.

Example:

Virgin PET = 69% of emissions

Meaning:

Best place to reduce emissions

Why Recycled Materials Reduce Emissions

Example:

Virgin PET = 2.15 kg CO₂e/kg

RPET = 0.55 kg CO₂e/kg

For 60,000 kg:

Virgin PET → 129 tCO₂e

RPET → 33 tCO₂e

Reduction:

≈96 tCO₂e

The factory isn't emitting less smoke.

It is reducing demand for carbon-intensive production upstream.

What Carbon Accounting Really Is

Not:

Counting carbon

But:

Tracking resources

↓

Finding emission hotspots

↓

Identifying better alternatives

↓

Quantifying impact

One-Line Mental Model

Resources In

↓

Industrial Activities

↓

Emissions Generated

↓

Leak Point Detection

↓

Circular Alternatives

↓

Cost & Carbon Savings

Your project isn't just a **carbon calculator**.

It's an **AI-powered industrial resource flow and circular economy advisor** that helps companies understand where emissions originate and what actions provide the biggest environmental and financial return.

Full Forms

Carbon & Sustainability

CO₂ (Carbon Dioxide) — Primary greenhouse gas released from burning fossil fuels.

CO₂e (Carbon Dioxide Equivalent) — Standard unit that converts all greenhouse gases into an equivalent amount of CO₂.

GHG (Greenhouse Gas) — Heat-trapping gases like CO₂, methane, and nitrous oxide.

Carbon Footprint — Total emissions caused directly and indirectly by an organization or activity.

tCO₂e (Tonnes of CO₂ Equivalent) — Carbon footprint reporting unit; 1 tCO₂e = 1000 kg CO₂e.

Emission Scopes

Scope 1 — Direct emissions from company-owned sources (diesel generators, LPG boilers, etc.).

Scope 2 — Indirect emissions from purchased electricity, steam, or heat.

Scope 3 — Emissions from the value chain, such as materials, transport, suppliers, and waste.

GHG Protocol — Global standard used to classify and calculate Scope 1, 2, and 3 emissions.

Circular Economy

CE (Circular Economy) — System focused on reducing waste through reuse, recycling, and recovery.

Closed-Loop Recycling — Waste is recycled back into the same production process.

Open-Loop Recycling — Waste is recycled into a different product or industry.

Circularity Score — Measure of how effectively resources are reused or recycled.

Plastics & Materials

PET (Polyethylene Terephthalate) — Common plastic used in bottles and packaging.

RPET (Recycled PET) — Recycled version of PET with significantly lower carbon footprint.

HDPE (High-Density Polyethylene) — Durable plastic used in containers, pipes, and bottles.

LDPE (Low-Density Polyethylene) — Flexible plastic used in films, bags, and packaging.

PP (Polypropylene) — Lightweight plastic used in packaging and consumer goods.

PVC (Polyvinyl Chloride) — Plastic widely used in construction and piping.

Energy & Utilities

LPG (Liquefied Petroleum Gas) — Common industrial fuel used for heating and boilers.

PNG (Piped Natural Gas) — Natural gas supplied through pipelines.

CNG (Compressed Natural Gas) — Compressed methane used as a cleaner fuel.

kWh (Kilowatt Hour) — Standard unit of electrical energy consumption.

MWh (Megawatt Hour) — 1000 kWh of energy.

GWh (Gigawatt Hour) — 1,000,000 kWh of energy.

Financial Metrics

CAPEX (Capital Expenditure) — One-time investment in assets or infrastructure.

OPEX (Operational Expenditure) — Recurring expenses required to run operations.

ROI (Return on Investment) — Measure of how much value an investment generates.

Payback Period — Time required to recover the initial investment cost.

AI & Data

AI (Artificial Intelligence) — Systems that perform tasks requiring human-like decision making.

ML (Machine Learning) — AI techniques that learn patterns from data.

LLM (Large Language Model) — AI model trained on massive text datasets for reasoning and language tasks.

OCR (Optical Character Recognition) — Technology that extracts text from images and PDFs.

API (Application Programming Interface) — Mechanism allowing different software systems to communicate.

ETL (Extract, Transform, Load) — Process of collecting, cleaning, and storing data for analysis.

Sustainability Reporting

ESG (Environmental, Social, Governance) — Framework used to assess corporate sustainability performance.

LCA (Life Cycle Assessment) — Analysis of environmental impacts across a product's entire lifecycle.

EPD (Environmental Product Declaration) — Verified document reporting a product's environmental footprint.

DQI (Data Quality Index) — Metric indicating reliability and completeness of collected data.

Project-Specific Terms

Leak Point — Activity contributing the largest share of total emissions.

Hotspot Analysis — Process of identifying major emission sources.

Emission Factor — Value used to convert an activity into emissions.

Material Substitution — Replacing a high-emission material with a lower-emission alternative.

Digital Twin — Virtual model used to simulate industrial processes and sustainability improvements.

What-If Analysis — Simulation of alternative scenarios before implementation.

Possible Exploit

The Scope 3 will be always high, so the solution would always be around changing the material.

Stuff to do

- simplify technical stuff
- give more suggestions with brief about what is causing the problem and how to fix it
- add more material option and a text box to write it too.
- transparently explain, how we reached to the conclusion.
- what-if stuff.
- GENERATING CPCB compliant report directly in the SaaS 🚀